

TAMPINES JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
[mark scheme]



CANDIDATE
NAME

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CIVICS GROUP

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TUTOR
NAME

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Biology

9648/01

Paper 1 Multiple Choice

Tuesday, 24 September 2013

1 hour 15 minutes

Additional Materials: Optical Answer Sheet (OAS)

READ THESE INSTRUCTIONS FIRST

Write your name and Civics Group in the spaces at the top of this page and on any separate writing paper used.

There are **40** questions in this paper.

Answer **all** questions. For each question there are four possible answers **A, B, C** and **D**.

Choose the **one** you consider correct and record your choice in **soft pencil** on the separate **OAS**.

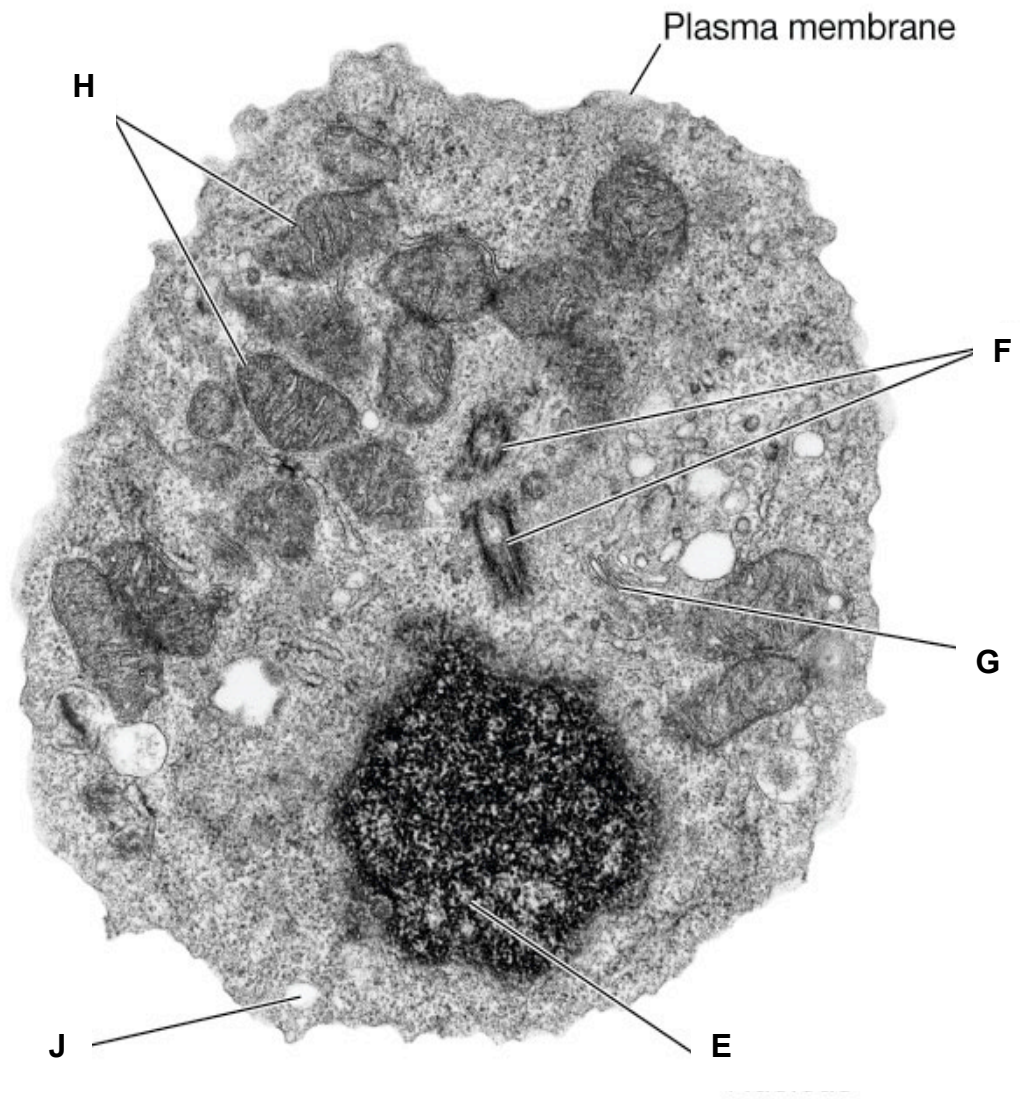
Read the instructions on the OAS very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.

Any rough working should be done in this booklet.

MULTIPLE CHOICE QUESTIONS (40 MARKS)

1. An electron micrograph of a cell is shown below.



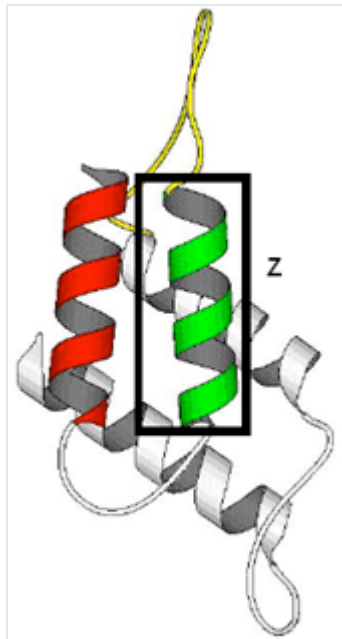
Match the organelles **E, F, G, H** and **J** associated with the cellular processes listed.

	E	F	G	H	J
A	DNA replication	Digestion of material	Organizes the spindle	Oxidative phosphorylation	Packaging of secretory products
B	Oxidative phosphorylation	Organizes the spindle	Digestion of material	DNA replication	Packaging of secretory products
C	Organizes the spindle	Digestion of material	Oxidative phosphorylation	Packaging of secretory products	DNA replication
D	DNA replication	Organizes the spindle	Packaging of secretory products	Oxidative phosphorylation	Digestion of material

2. Which row correctly links a feature of a phospholipid molecule with its function in membranes?

	Both polar and non-polar (amphipatic)	Phosphate group	Unsaturated fatty acids	Saturated fatty acids
A	Allowing diffusion of small molecules	Aiding membrane stability	Decreasing fluidity	Binding proteins
B	Forming a bilayer	Repelling charged molecules	Increasing fluidity	Decreasing fluidity
C	Forming glycolipids	Binding proteins	Increasing permeability	Increasing fluidity
D	Forming vesicles	Forming glycolipids	Binding proteins	Decreasing permeability

3. The diagram below shows the structure of a general transcription factor.



Which correctly describes the structure of the part enclosed in box Z?

	<u>Number of Amino Acids Present</u>	<u>Types of bonds Present</u>
A	Cannot be determined	Hydrogen Bonds only
B	Cannot be determined	Hydrogen Bonds and hydrophilic bonds
C	Approximately 11 amino acids	Hydrogen bonds only
D	Approximately 11 amino acids	Hydrogen bonds and hydrophobic bonds

4. Each polypeptide chain of haemoglobin contains a number of regions in the form of an alpha helix.

Which feature of the haemoglobin molecule is responsible for this?

- A bonding between the four polypeptide chains
- B hydrophobic interactions at the centre of the molecule
- C the amino acid sequence of the globin
- D the presence of iron in the haem groups

5. Which of the following statement about enzymes is inaccurate?

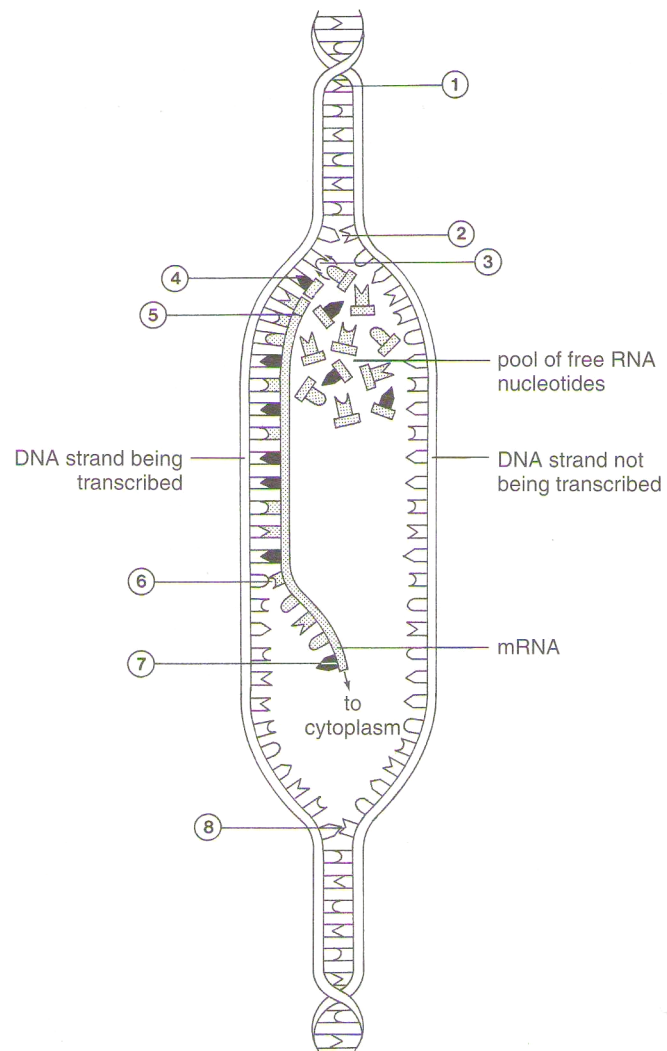
- A Enzyme function is not influenced by factors such as lead nitrate.
- B Enzyme function is reduced if the three-dimensional structure or conformation of an enzyme is altered.
- C Enzymes increase the rate of chemical reaction by lowering activation energy barriers.
- D Enzymes may require a non-protein cofactor or ion for catalysis to take place.

6. Which of the following does not describe the active site of an enzyme?

- A Its specificity is defined by the arrangement of the monomers.
- B It is usually a crevice or cleft.
- C It initially binds substrates by weak attractions.
- D It is two-dimensional in structure.

7.

5



Which of the following statements is true?

- A The bond in 1 is phosphodiester bond.
- B The bond in 5 is peptide bond.
- C 7 refers to the 5' end of the mRNA.**
- D 8 is showing unwinding of the DNA molecule.

8. The following coding sequence of DNA is taken randomly from a bacterial genome.

3' TTACGCTTCGAAATAGGAATATCATAGGCT 5'

This sequence is cloned into a plasmid and transformed into a suitable host. What would be the first four amino acids of a polypeptide generated from this sequence as expressed by the host?

The mRNA codons for some amino acids are shown in the table below:

Arg	CGA, CGG, AGA, AGG	Leu	CUU, CUC, CUA, CUG
Asp	GAU, GAC	Lys	AAA, AAG
Ile	AUU, AUC, AUA	Phe	UUU, UUC
Start	AUG	Ser	UCA, UCG, AGU, AGC
Stop	UAG, UGA, UAA	Tyr	UAU, UAC

- A** Met-Arg-Ser-Phe
- B** Met-Arg-Ser-Lys
- C** Met-Ile-Phe-Leu
- D** Met-Tyr-Lys-Asp.
9. Which of the following is unique to mitosis and not a part of meiosis?
- A** Homologous chromosomes pair forming bivalent.
- B** Homologous chromosomes cross over.
- C** Chromatids are separated during anaphase.
- D** Homologous chromosomes behave independently.
10. Cells treated with colchicine often mutate. They fail to produce spindles during cell division. What would be the result of this?
- A** chromosome deletion
- B** chromosome translocation
- C** gene deletion
- D** polyploidy

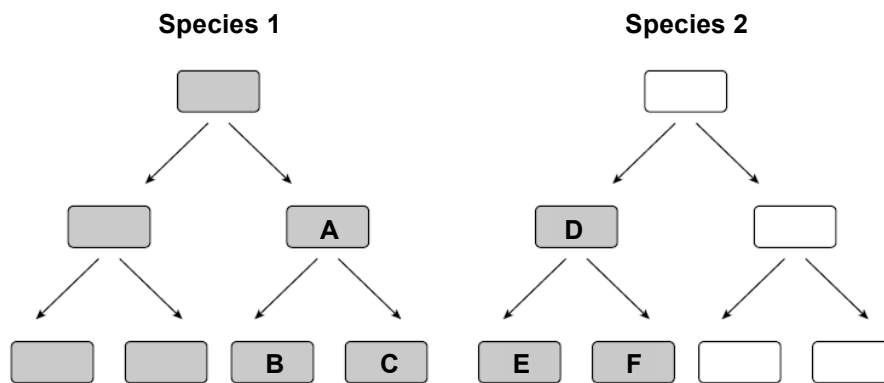
11. Four bacteriophage genes were mapped by recombination. The recombination frequencies between the four genes are shown in the table below.

	L	M	N	P
L	--	0.18	0.10	0.20
M		--	0.15	0.37
N			--	0.24
P				--

What is the order of the genes along the chromosome?

- A** L, M, N, P
B N, M, P, L
C M, L, P, N
D P, L, N, M
12. During a viral infection, attachment is usually specific to a particular cell type because
- A** the virus is attracted to the appropriate host cells by proteins secreted into the extracellular fluid.
B the virus recognises and binds to specific molecules in the cytoplasm of the host cell.
C the virus recognises and binds to specific molecules on the surface of the host cell.
D the host cell produces channel proteins that provide passageways for the viruses to enter the cytoplasm.
13. When glucose is absent and lactose is present, lactose binds to the operon's repressor. Which statement best describes what happens subsequently?
- A** Repressor binds to promoter, transcription is initiated.
B Repressor binds to operator, transcription is not initiated.
C Repressor fails to bind to structural genes, transcription is not initiated.
D Repressor fails to bind to the operator, transcription is initiated

14. The diagram below shows 2 species of bacteria. The bacteria that are shaded are resistant to the antibiotic penicillin.



Which one of the following statements is likely to be true?

- A** Bacterium **D** is resistant to penicillin due the transfer of the complete F plasmid and penicillin resistant gene through transduction involving Bacterium **A**.
- B** Bacterium **D** is resistant to penicillin due to horizontal gene transfer involving a bacteriophage and Bacterium **A**.
- C** Bacterium **E** acquired resistance to penicillin as a result of random mutation and genetic transformation.
- D** Bacteria **E** and **F** are resistant to penicillin as a result of conjugation from Bacterium **D**.
15. The statements are descriptions of aspects of genetics.
1. The phenotype is affected by both alleles at the same locus of a heterozygous individual.
 2. The combined effects of alleles at two or more gene loci equal the sum of their individual effects.
 3. Many different alleles present in a gene pool can occupy the same gene locus.
 4. Alleles of one gene mask the effects of the alleles of another gene at a different locus.

Match the description to each of the following terms

	codominant alleles	epistatic alleles	multiple alleles	additive genes
A	1	4	2	3
B	1	4	3	2
C	2	1	4	3
D	3	1	4	2

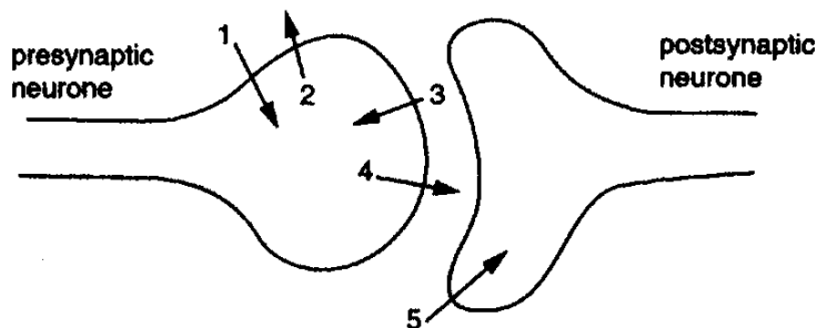
16. Skin colour in a variety of pumpkins is controlled by two pairs of alleles, Pp and Rr, which segregate independently.

The allele P is dominant and must be present for the development of pigmentation in the skin. Pumpkins without pigment are white. Allele R is dominant and produces a red pigment in the skin. The recessive allele r gives a yellow colour.

In a testcross with a parent of unknown genotype, a phenotypic ratio of 1 red : 1 white was obtained.

What was the genotype of the unknown parent?

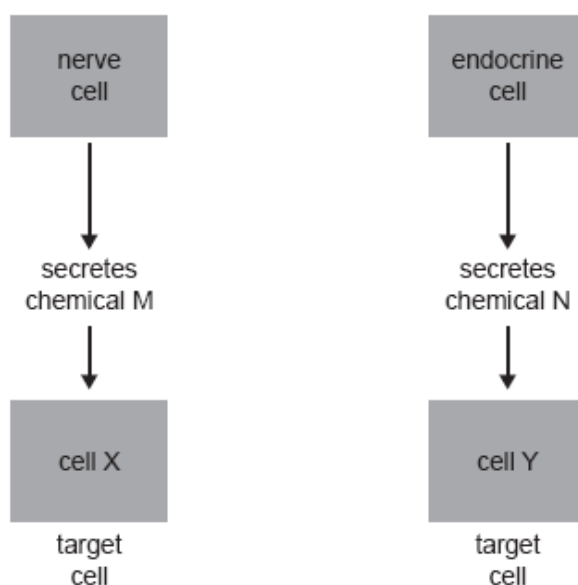
- A PPRR
- B PpRR**
- C PPRr
- D Pprr
17. Studies on the giant axon of the squid showed that bacteria toxin BAXIN results in an increase in negativity of the postsynaptic membrane. What is the possible mode of action of BAXIN?
- A The release of acetylcholine
- B Analog of acetylcholine
- C Release of potassium ions from postsynaptic membrane**
- D Mimicking of cholinesterase
18. The diagram shows the sequence of events occurring as an action potential arrives at the synapse. The numbered arrows represent the movement of substances across the membranes.



What are the substances moving across the membranes?

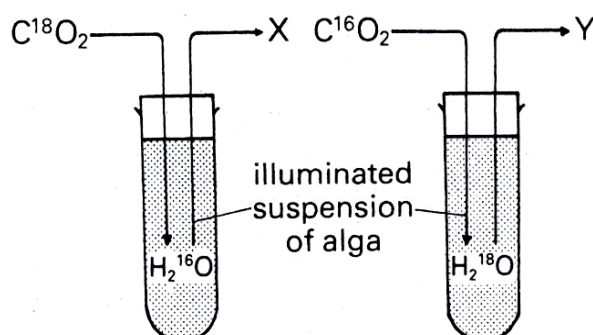
	1	2	3	4	5
A	K^+	Na^+	acetylcholine	Ca^{2+}	K^+
B	K^+	Na^+	Ca^{2+}	acetylcholinesterase	acetylcholine
C	Na^+	K^+	Ca^{2+}	acetylcholine	Na^+
D	Na^+	K^+	acetylcholinesterase	acetylcholine	Ca^{2+}

19. The diagram shows how a nerve cell and an endocrine cell interact with cell X and cell Y respectively.



Which of the following is true of both interactions?

- A** It involves the movement of chemicals M and N to their respective target cells.
- B** Chemical M is secreted faster than chemical N.
- C** When both cells X and Y are stimulated, phosphorylation cascades must be initiated.
- D** Both chemicals M and N must bind to cell-surface receptors on cell X and Y respectively.
20. In the experiment shown in the diagram, the oxygen given off at X and Y will be labelled with the isotopes _____ and _____ respectively.



	X	Y
A	^{16}O	^{16}O
B	^{16}O	^{18}O
C	^{18}O	^{16}O
D	^{18}O	^{18}O

21. The weed killer DCMU blocks the flow of electrons from the electron transport chains in photophosphorylation.

Why does this kill the plant?

- A Active transport of mineral ions is prevented.
- B ATP and reduced NADP are not produced.**
- C Photoactivation of the chlorophyll cannot occur.
- D Photolysis of water does not occur.

22. To what extent can glucose be oxidised by the soluble portion of cytoplasm of muscle cells, from which all organelles have been removed?

- A Acetyl-CoA but no further
- B Carbon dioxide and water
- C Lactate but no further
- D Pyruvate but no further**

23. In which of the following changes is most energy released during the metabolism of carbohydrates?

- A Maltose to glucose
- B Glucose to phosphoglyceraldehyde
- C Phosphoglyceraldehyde to pyruvic acid
- D Pyruvic acid to carbon dioxide and water**

24. Which of the following is **true** of gene regulation in both prokaryotes and eukaryotes?

- A Gene regulation of eukaryotes involves bending of DNA while in prokaryotes, chemical modifications to DNA is required.
- B In eukaryotes, increase in efficiency of RNA polymerase requires specific transcription factors to activate the initiation complex while in prokaryotes, activators are required to bind directly to promoter site to increase efficiency of RNA polymerase.**
- C In eukaryotes, RNA polymerase binds to the TATA box while in prokaryotes, RNA polymerase binds to the regulatory gene.
- D Repressors and activators regulate expression in eukaryotes while prokaryotes are regulated by repressors only.

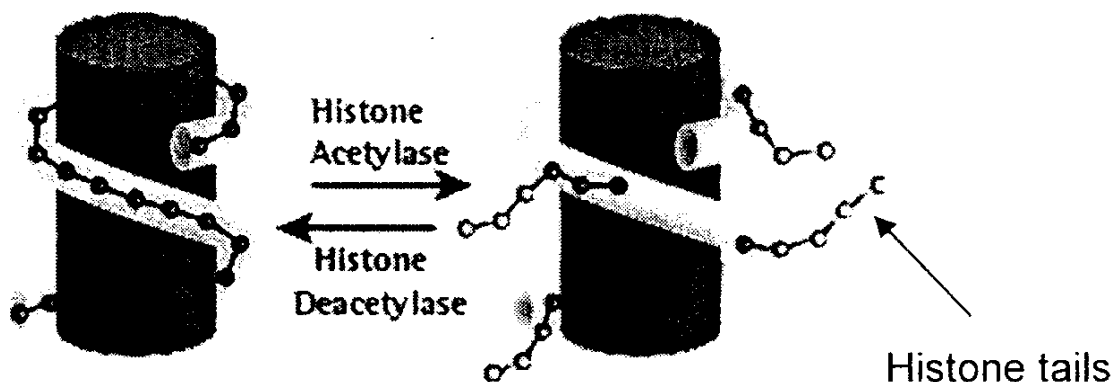
25. The following events occur in the extension of telomeres.

- 1 Further extension of 3' end of telomere.
- 2 Translocation of telomerase to end of telomere.
- 3 Reverse transcription to extend the 3' end of the telomere.
- 4 Complementary base-pairing of RNA template with single-stranded end of telomere.
- 5 DNA polymerase catalyses formation of complementary strand to form double-stranded DNA.

In which order do these events take place?

- A 3 2 4 5 1
 B 3 4 2 1 5
 C 4 2 3 5 1
 D 4 3 2 1 5

26. As shown in the diagram below, acetylation of histones promotes loose chromatin structure. Recent evidence has shown that chemical modification of histones play a direct role in regulation of gene expression. Suggest how this works.



- A Helicase action is enhanced by acetylation.
 B Acetylation of histones neutralizes their negative charges and encourages binding to DNA polymerase.
 C When nucleosomes are highly acetylated, chromatin becomes less compact and DNA is more accessible for transcription.
 D RNA polymerase works better by binding with acetyl groups.

27. The hormone insulin binds to the tyrosine kinase receptors and initiates various signal transduction pathway to generate cellular responses. Which of the following shows the

correct sequence of events, following the binding of insulin to the receptor?

- a. phosphorylation of tyrosine kinases
- b. dimerization of tyrosine kinase receptor
- c. activation of enzyme

- A** a,b,c
- B** c,b,a
- C** b,a,c
- D** c,a,b

28. Many signal transduction pathways use second messengers to

- A** amplify the message by directly phosphorylating proteins.
- B** induce a signal from the outside to the inside of the cell.
- C** relay the message from the inside of the membrane throughout the cytoplasm.
- D** transport a signal through the lipid bilayer portion of the plasma membrane.

29. Which of the following is a similarity between G-protein-linked receptors and tyrosine-kinase receptors?

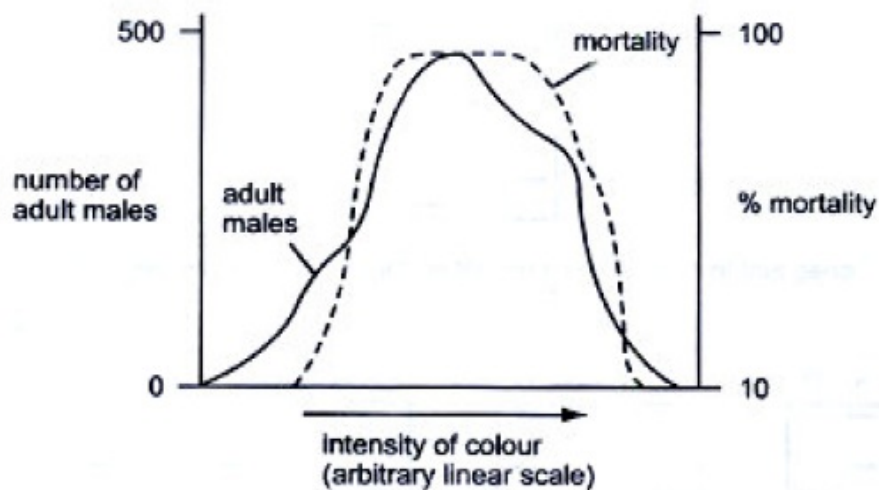
- A** activation that results from binding of GTP
- B** formation of a dimer following binding of a signal molecule
- C** regions of the receptor that span the plasma membrane
- D** signal-binding sites specific for steroid hormones

30. Which of the following increases the number of different alleles in a population?

- A** crossing over
- B** gene mutation
- C** random fusion of gametes
- D** random assortment of chromosomes in meiosis

31. The graph below shows data on a population of a species of moth which shows

considerable variation in colour intensity. Which conclusion can be made from this graph?



- A Colour variation is environmentally induced.
- B Colour variation is genetically determined.
- C Extreme forms are favoured by natural selection.**
- D The species shows discontinuous variation with respect to colour.

32. Approximately 1 in 20 Europeans are heterozygous for a recessive allele responsible for the genetic condition, cystic fibrosis (CF). People who are homozygous for CF have a reduced life expectancy. Heterozygotes are more resistant to some bacterial infections of the gut, such as typhoid fever, than homozygotes for the normal, dominant allele.

What could explain the high incidence of the recessive CF allele in the European population?

- A natural selection favouring heterozygotes**
- B natural selection favouring homozygotes for the recessive CF allele
- C lack of genetic drift in the European population
- D Mutation

33. What is one similarity and one difference between classification and phylogeny?

- A** Both organise organisms into groups based on their characteristics but phylogeny takes into consideration the evolutionary relationships between groups.
- B** Both put organisms together according to increasingly inclusive groups but classification does not show a branching pattern of evolutionary relationships.
- C** Both rely on physical characteristics of organisms but phylogeny does not require scientific names.
- D** Both take into account the common ancestors of organisms but classification is hierarchical in nature.

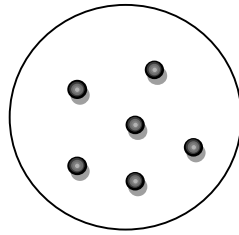
34. Which uses of the information from the human genome project are generally considered to be unethical?

1. an insurance company only giving cheap rates to people with genetic predispositions to fewer diseases
2. genetic archaeologists identifying the earliest forms of genes to show evolutionary relationships
3. cytologists developing tests for only some defective genes
4. doctors only giving specific drugs to block the actions of faulty genes to carriers of those genes
5. genetic counselors giving specific lifestyle information only to people genetically predisposed to risks
6. parents choosing embryos for implantation only after ante-natal tests for acceptable genes

A	1 and 3
B	1 and 6
C	2 and 5
D	3 and 4

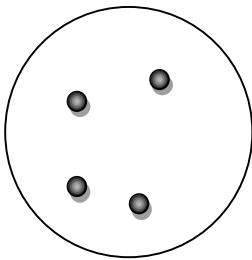
35. The insulin gene and a plasmid with ampicillin and tetracycline resistant genes were cut

using the same restriction enzyme, *EcoRI*. The cut plasmid and insulin gene was mixed together and DNA ligase was allowed to react. Subsequently the mixture was heat shocked at 42 degrees with *E. coli*. The mixture was then plated on nutrient agar to obtain a master plate with colonies shown below.

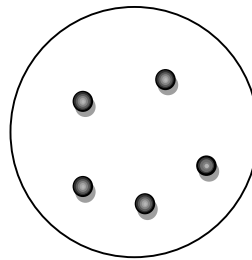


Master plate (nutrient agar)

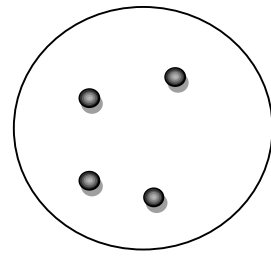
Three replica plates containing different antibiotics were obtained from the master plate. The diagram below shows the results of the replica plates.



**Nutrient agar with
ampicillin**



**Nutrient agar with
tetracycline**



**Nutrient agar with
tetracycline and
ampicillin**

Which of the following can be concluded from the results obtained?

A	The ampicillin resistant gene is inactivated by insertion of the insulin gene
B	The tetracycline resistant gene is inactivated by insertion of the insulin gene.
C	The tetracycline resistant gene does not act as a selectable marker
D	The ampicillin resistant gene does not act as a selectable marker.

36. A student performed an experiment to determine if a particular gene has been inserted in

a genetically modified organism.

- i. Transfer of DNA to nitrocellulose membrane
- ii. Restriction digestion of genomic DNA
- iii. Cleaved DNA separated using gel electrophoresis
- iv. Create radioactive probe
- v. Incubate probe and membrane

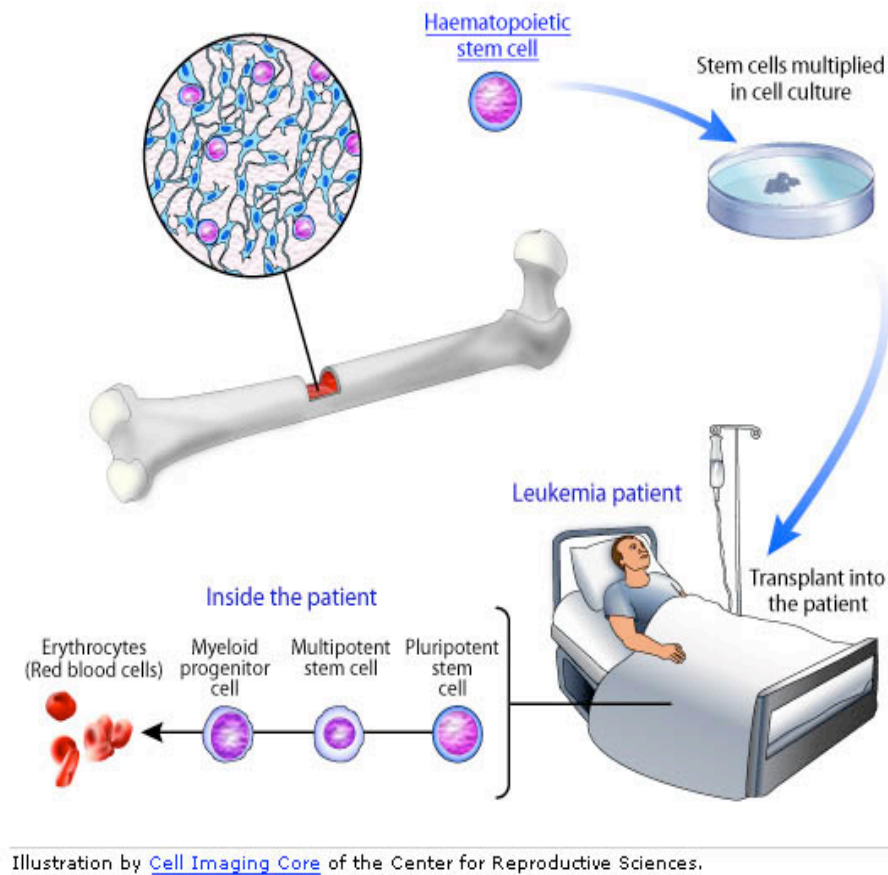
Which is the correct sequence to the above steps?

A	ii → iii → i → v → iv.
B	ii → iii → i → iv → v.
C	iv → v → i → ii → iii.
D	iii → i → ii → iv → v.

37. The genetically engineered super salmon was created from Atlantic salmon stocks and are capable of growing to a large size in 14 months. Which of the following is not a benefit intended from the crop?

- A** Higher yield for farmers
- B** Minimising pollution
- C** Decreasing the food consumption of the crops in their lifetime
- D** Increase in supply to meet world's demand

38.



Which of the following options is not a characteristic of the form of treatment shown above?

- 1 Easy to extract in laboratory
- 2 Readily available
- 3 Difficult to control
- 4 Limited longevity

A 1 only B 1 and 2 C 1 and 3 D 1, 2, 3 and 4

39. Germ-line gene therapy has not been prevalent in medical treatment because of many reasons. Which of the following is/are implications of germ-line therapy?

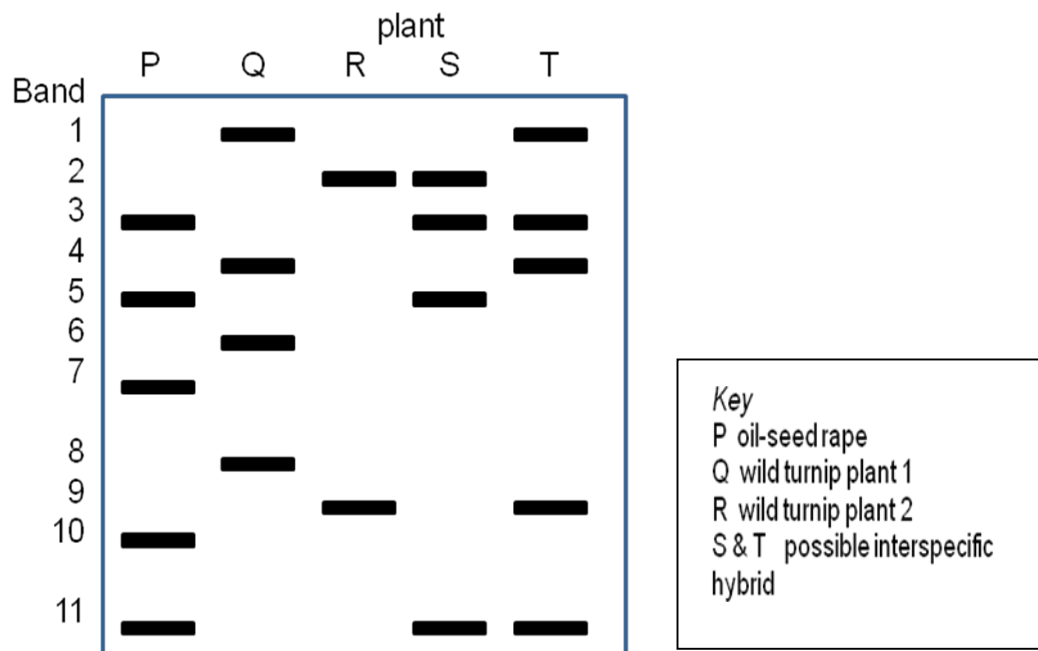
- 1 unforeseeable clinical risks
- 2 opens problems for social discrimination
- 3 cost
- 4 issue of consent

A 1 only B 2, 3 and 4 C 1, 2 and 4 D 1, 2, 3 and 4

40. A possible route of escape of an inserted gene from a genetically engineered crop of oil-seed rape, *Brassica napus*, is into populations of the wild turnip, *Brassica rapa*. Two populations of wild turnip growing next to large fields of a cultivar of oil-seed rape were studied. Seeds were collected from these wild turnips plants and their DNA analysed using restriction enzymes and electrophoresis.

Using the key provided below, determine

- whether plants S and/or T are interspecific hybrids and
- the parental plants they are derived from



	<i>Interspecific hybrid</i>	<i>Parent plants</i>
A	Both S and T	S: P x R T: P x Q
B	Both S and T	S: P x Q T: P x R
C	T only	Q x R
D	S only	P x R